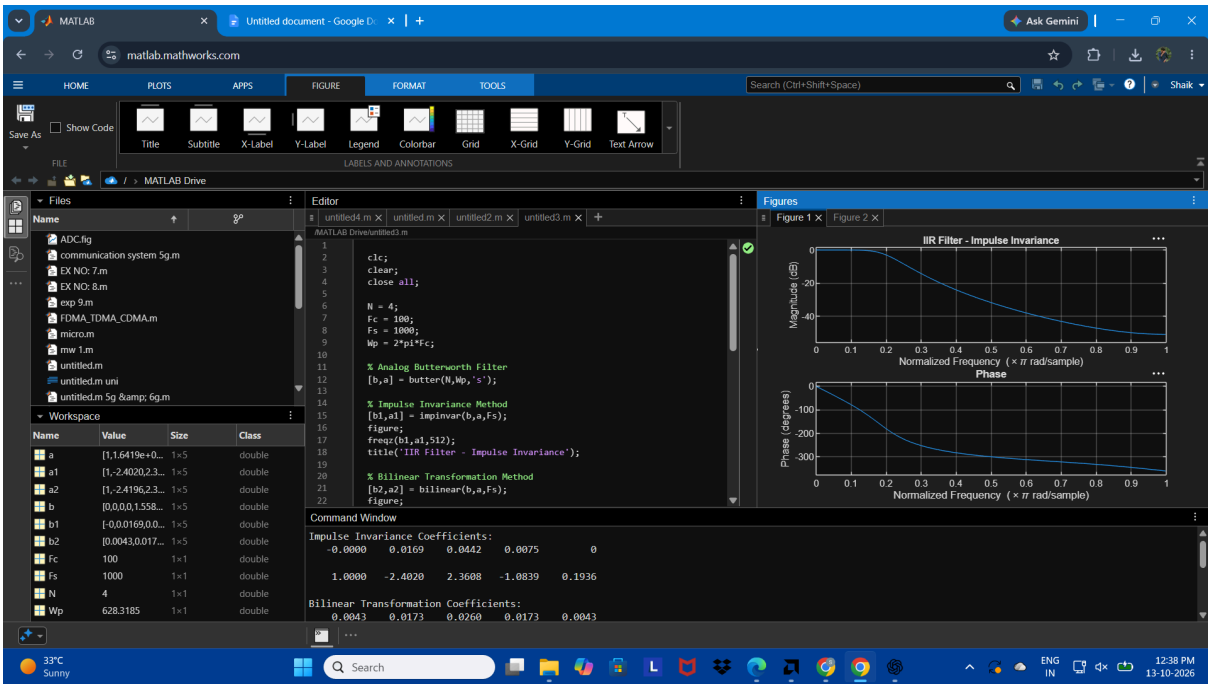
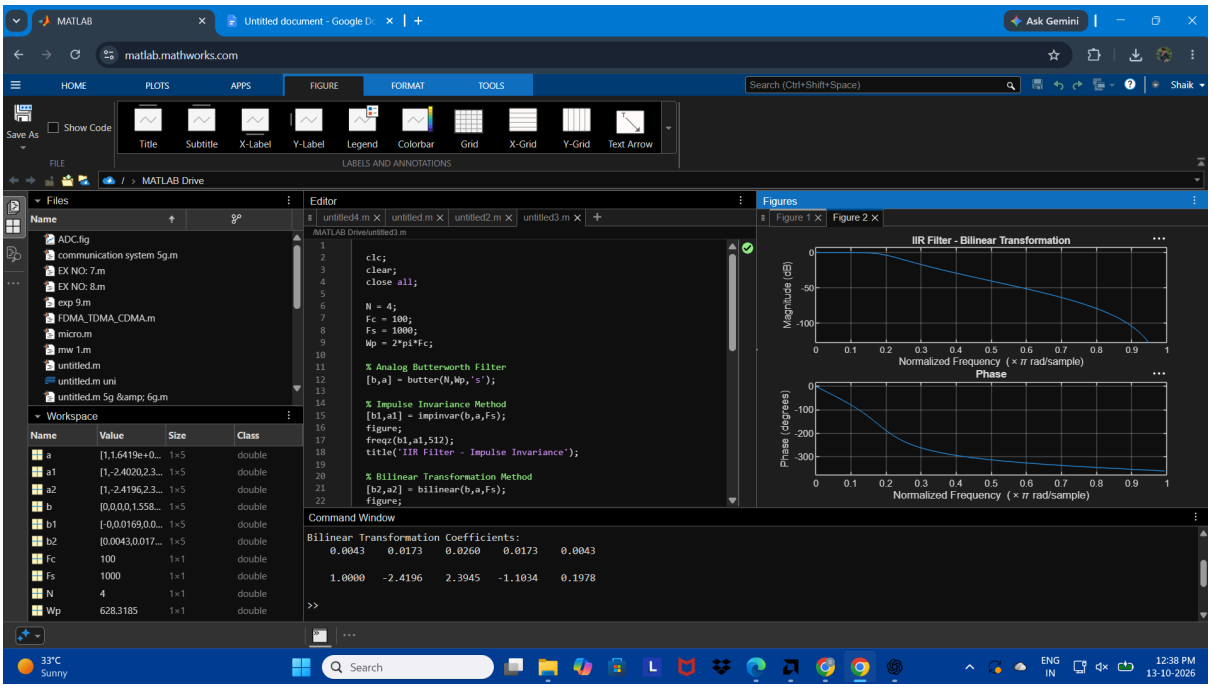
EXP NO:- 3

PROGRAM ➖

```
clc;
clear;
close all;
N = 4;
Fc = 100;
Fs = 1000;
Wp = 2*pi*Fc;
% Analog Butterworth Filter
[b,a] = butter(N,Wp,'s');
% Impulse Invariance Method
[b1,a1] = impinvar(b,a,Fs);
figure;
freqz(b1,a1,512);
title('IIR Filter - Impulse Invariance');
% Bilinear Transformation Method
[b2,a2] = bilinear(b,a,Fs);
figure;
freqz(b2,a2,512);
title('IIR Filter - Bilinear Transformation');
disp('Impulse Invariance Coefficients:');
disp(b1);
disp(a1);
disp('Bilinear Transformation Coefficients:');
disp(b2);
disp(a2);
```

OUTPUT:-


IIR Filter - Impulse Invariance
Magnitude (dB)
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
Normalized Frequency (× π rad/sample)
Impulse Invariance Coefficients:
-0.0000 0.0169 0.0442 0.0075 0
1.0000 -2.4020 2.3608 -1.0839 0.1936
Bilinear Transformation Coefficients:
0.0043 0.0173 0.0260 0.0173 0.0043


IIR Filter - Bilinear Transformation
Magnitude (dB)
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
Normalized Frequency (× π rad/sample)
Bilinear Transformation Coefficients:
0.0043 0.0173 0.0260 0.0173 0.0043
1.0000 -2.4196 2.3945 -1.1034 0.1978